Attitude predicates

Day 3: Nominals and the event semantics approach

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- the proposition *that Nynke is going to be hired* is combining with that NP/entity.

Hintikka semantics is *tacit* about this, we need to refine it (not shoot it out of the sky).

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The discussion here is based on languages not making a distinction.

The basic idea

One way for a uniform semantics across *believe NP* and *believe CP* is

- $Q1$ to say that *believe* only combines with entities,
- $Q2$ and to propose a way of combining NPs with clauses.

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(3) a. Jones buttered the toast in the bathroom with the knife at midnight. (Davidson 1967)

b. $\exists e[\textsc{butter}(e) \wedge \textsc{agent}(e, \text{jones}) \wedge \textsc{patient}(e, \text{the toast}) \wedge \textsc{in}(e, \text{the bathroom}) \wedge \textsc{at}(e, \text{midnight})]$

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For *Jort believes the rumor*, you might then get:

(4) $\exists e[\textsc{believe}(e) \wedge \textsc{experiencer}(e, \text{jort}) \wedge \textsc{theme}(e, \text{the rumor})]$

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Notes:

- If you worry about events vs. states, here we bundle the two together, but see Krifka et al. (1995).
- If you wanted to decompose *the rumor* further, you could (exercise on your own).

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This will now require us to update our semantics for *that* clauses.

A proposal for $Q2$: Contents of entities

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(5) [The rumor/story/fact] is [that Ebenezer and Balthasar are going to break up].

We can leverage this idea to account for the fact that many predicates can take both nominal and clausal arguments!

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This is basically the template of any verb in Neo-Davidsonian semantics.

(7) $[\![\text{defenestrate}]\!] = \lambda e_v.defenstrate'(e)$

Arguments of verbs are introduced in the syntax by heads which contribute thematic role functions.

(8) a. Joan defenestrated Marge.
b. $\exists e[defenestrate'(e) \wedge \textsc{agent}(e) = j \wedge \textsc{theme}(e) = m]$

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(Kratzer 2006, Moulton 2009, 2015, Bondarenko & Elliott 2026)

'the set of entities/events whose content is a subset of that of the proposition *S*'

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Flexible typing: allow for composition with both nominals (*rumor*) and verbs (*believe*)

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Solution: Composition by **predicate modification** (invented to handle modifiers, stuff like adjective + noun composition)

Predicate modification (informal-ish)

We can combine two expressions of type $\langle \alpha, t \rangle$, where α is any type, to create a new expression of type $\langle \alpha, t \rangle$ whose meaning is a conjunction of those two expressions.

The resulting composition for *(the) rumor that*

We can 'predicate modify' the denotation of *rumor* and the denotation of the *that* clause.

(10) a. rumor that Nynke is going to be hired

b. $\lambda x.rumor(x)$

c. $\lambda x.rumor(x) \wedge$
$\textsc{Cont}(x) \subseteq \{w' \mid [\![\text{Nynke is going to be hired}]\!]^{w'} = 1\}$

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We can then apply whatever operation we can apply to NPs to *rumor that p*, e.g., the definite article THE or existential closure (below).

(11) a. the rumor that Nynke is going to be hired

b. THE *x such that* $rumor(x) \wedge$
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The resulting composition for *believe the rumor that*

It 'automatically follows' what the composition would look like for *believe the rumor that p*.

(12) a. Jort believes the rumor that Nynke is going to be hired

b. $\exists e.\textsc{believe}(e) \wedge \textsc{experiencer}(e, \text{jort}) \wedge \textsc{theme}(e, \textsc{the}\ x\ \textit{such that rumor}(x) \wedge \textsc{Cont}(x) \subseteq \{w' \mid [\![\text{Nynke is going to be hired}]\!]^{w'} = 1\})$

The resulting composition for *believe that*

Kratzer and others originally proposed (simplified) that *believe that p* involved a null nominal direct object:

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$\approx$ Jort believes **something whose content is that** Nynke is going to be hired.

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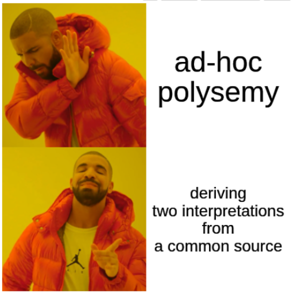
We would thus get, in keeping with what we saw about *the rumor that p* above, modulo existential closure:

(14) Jort believes that Nynke is going to be hired.

$$\exists e \exists x.\textsc{believe}(e) \wedge \textsc{experiencer}(e, \text{jort}) \wedge \textsc{theme}(e, x) \wedge \textsc{Cont}(x) \subseteq \{w' \mid [\![\text{Nynke is going to be hired}]\!]^{w'} = 1$$

What did we get for all this?



- A common *believe* which applies to both nominal and declarative complements
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- The right entailments from attitude + nominal to attitude + clause sentences
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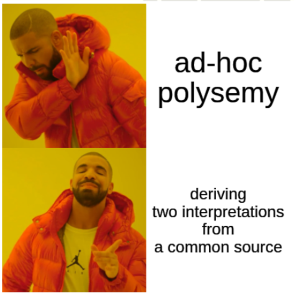

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*What about *want*...?

Events?

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But we do need to incorporate events into the semantics of attitude reports,

- based on at least two pieces of evidence, this justifies part of the semantics given above,
- and it leads to a refinement as to *what it is exactly* whose content is being specified, a null object? or an event…?

Piece of evidence #1: Explanans vs. explandum

(15) Jort explained that Nynke was going to be hired.
$\approx$ Jort said that Nynke was going to be hired (as an explanation for something else).

(16) Jort explained the rumor that Nynke was going to be hired.
$\approx$ Jort said something that explains the rumor Nynke was going to be hired.

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These are respectively called the ‘explanans’ and the ‘explanandum’ reading…

- explanans = the words or the thing that does the explaining
- explanandum = the thing to be explained

Piece of evidence #2: Aspect/question embedding

Another piece of evidence, not discussed today:

(17) a. */#Jort thinks whether Nynke is going to be hired.
  b. Jort is (actively) thinking whether Nynke is going to be hired.

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Events can have contents as well, just like other objects.

Capturing explanans vs. explanandum

Elliott (2017):

(18) a. Jort explained the rumor that Nynke was going to be hired.

b. $\exists e.\textsc{explain}(e) \wedge \textsc{experiencer}(e, \text{jort}) \wedge \textsc{theme}(e, \textsc{the}\ x\ \textit{such that rumor}(x) \wedge \textsc{Cont}(x) \subseteq \{w' \mid [\![\text{Nynke is going to be hired}]\!]^{w'} = 1\})$

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'J explained something by saying that N is going to be hired'

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But we find that you can say ‘x believes the rumor that p’, motivating the need for being able to combine propositions with nouns.

But also, you find meaning differences between things like ‘x explained that p’ vs. ‘x explained the rumor that p’.

Problem?

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- An independent and well motivated set of facts lead us to believe that attitude verbs describe eventualities, ones in which individuals are related to entities (*Mary believes something* $\approx$ *Mary kicked something*).
- This originally motivates the possibility of packing all of Hintikka's semantics into specifying the content of silent nouns combining with attitude verbs as their direct object.
- But could we just anchor clauses to event arguments as well, without going through silent nominals? (Many recent authors: **yes** (e.g. Roberts 2021, Djärv 2021, Bondarenko & Elliott 2026))

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Tomorrow & Friday: More work on these frontiers.

To be determined

In the meantime, several facts yet remain unexplained:

- Clausal-embedding predicates which don't embed nominals (**I think the rumor*)
- Embedding non-content nominals (*I know her*, *I believe the lifeguard*, see Djärv (2021, 2023)).
- Not all nominals behave the same (s/o Serra):

(20) a. the rumor that Gertrude was the murderer
     b. the proof that Gertrude was the murderer